

DISASTER RESOURCE BENCHMARK REPORT

Head-to-Head Comparative Study: TypeSafe JEV vs. Groq LLM (gpt-oss-120b)
Protocol: Dev/Test Split (Dev N=45, Test N=105) | Optimal JEV Threshold $\tau^* = 0.15$ | Date: 2026-10-05 20:12

EXECUTIVE BENCHMARK FINDINGS

Empirical evaluation shows Groq LLM achieved higher precision (0.892 vs 0.884, FPR: 2.2% vs 2.3%), while TypeSafe JEV achieved comparable recall (0.765 vs 0.782). Reliability: Both systems achieved 100% schema compliance (0 parse errors). TypeSafe JEV demonstrated an operational advantage in median latency (331ms vs 1331ms) and estimated cost (\$0.0390 vs \$0.2212/1k). Exact-match difference is not statistically significant (McNemar $p = 1.0000$).
Category Comparison: TypeSafe JEV Leads: 1 | Groq LLM Leads: 1 | Neutral Ties: 3 (out of 5 categories)

Category-by-Category Comparative Scorecard

Category	TypeSafe JEV	Groq LLM (120B)	Category Leader / Status
Classification Performance	Acc: 93.7% Rec: 0.765 Prec: 0.884 F1: 0.820	Acc: 94.1% Rec: 0.782 Prec: 0.892 F1: 0.833	Groq LLM (Higher F1)
Reliability & Hallucinations	Failures: 0 Invalid Keys: 0	Failures: 0 Invalid Keys: 0	TIE (Both 100% Reliable)
Ranking & Calibration	Micro AP: 0.869 AUC: 0.938 Brier: 0.057	Micro AP: 0.829 AUC: 0.916 Brier: 0.054	TIE
Operational Efficiency & Cost	p50: 331ms Cost/1k: \$0.0390	p50: 1331ms Cost/1k: \$0.2212	TypeSafe JEV (Faster & Cheaper)
Statistical Significance	Non-inf (Rec): False (margin=0.03) Prec: False	McNemar: No significant difference (p=1.0000)	TIE (No Significant Difference, $p \geq 0.05$)

DISASTER RESOURCE CLASSIFICATION BENCHMARK

TypeSafe JEV (OpenRouter Decisions API) vs. Groq LLM (gpt-oss-120b)

[HEAD-TO-HEAD BENCHMARK OVERVIEW]

Leading Dimension: Empirical Trade-off (Tied Categories)

JEV Leads: 1 Categories
LLM Leads: 1 Categories
Ties: 3 Categories

1. Classification Quality

TypeSafe JEV : Acc: 93.7% | Rec: 0.765 | Prec: 0.884 | F1: 0.820
Groq LLM : Acc: 94.1% | Rec: 0.782 | Prec: 0.892 | F1: 0.833

STATUS: Groq LLM (Higher F1)
Margin: Delta Prec: -0.008 | Delta Rec: -0.017

2. Reliability & Hallucinations

TypeSafe JEV : Failures: 0 | Invalid Keys: 0
Groq LLM : Failures: 0 | Invalid Keys: 0

STATUS: TIE (Both 100% Reliable)
Margin: Identical 100% Schema Adherence (0 errors)

3. Ranking & Calibration

TypeSafe JEV : Micro AP: 0.869 | AUC: 0.938 | Brier: 0.057
Groq LLM : Micro AP: 0.829 | AUC: 0.916 | Brier: 0.054

STATUS: TIE
Margin: ECE: 0.040 vs 0.021

4. Efficiency & Operations

TypeSafe JEV : p50: 331ms | Cost/1k: \$0.0390
Groq LLM : p50: 1331ms | Cost/1k: \$0.2212

STATUS: TypeSafe JEV (Faster & Cheaper)
Margin: Speedup: 4.0x | Cost: 5.7x Cheaper

5. Statistical Rigor

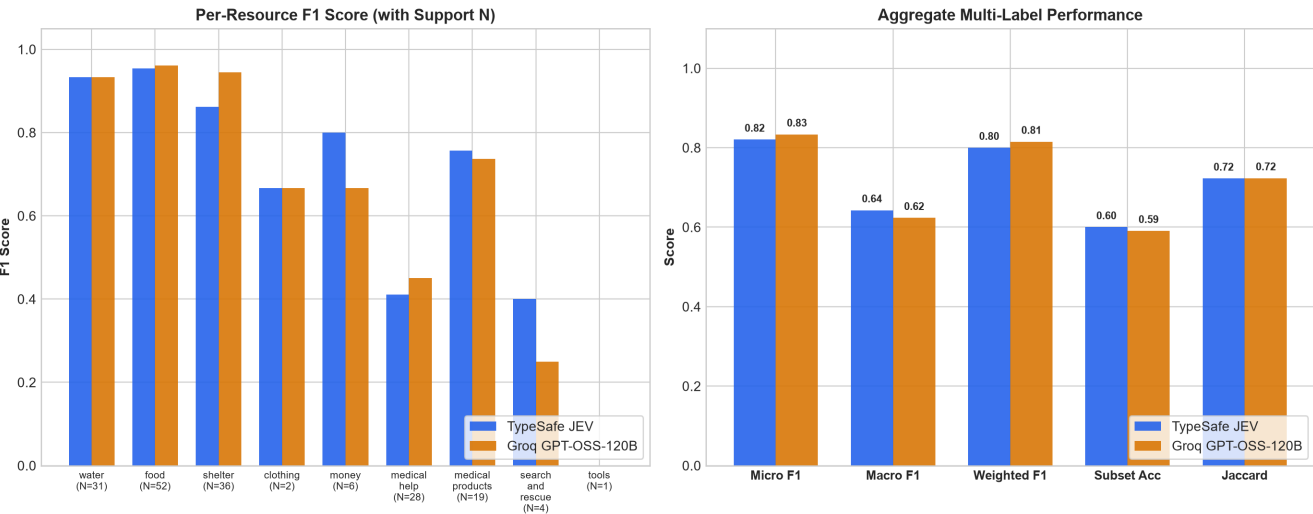
TypeSafe JEV : Non-Inf (Rec): False (margin=0.03) | Prec: False
Groq LLM : McNemar: No significant difference (p=1.0000)

STATUS: TIE (No Significant Difference, p >= 0.05)
Margin: Rec p=0.273 | Prec p=0.185

Multi-Label Classification Evaluation (Test Set)

Complete multi-label metrics evaluated on the test set. Headline numbers reflect optimal dev-tuned threshold (tau* = 0.15) for TypeSafe JEV.

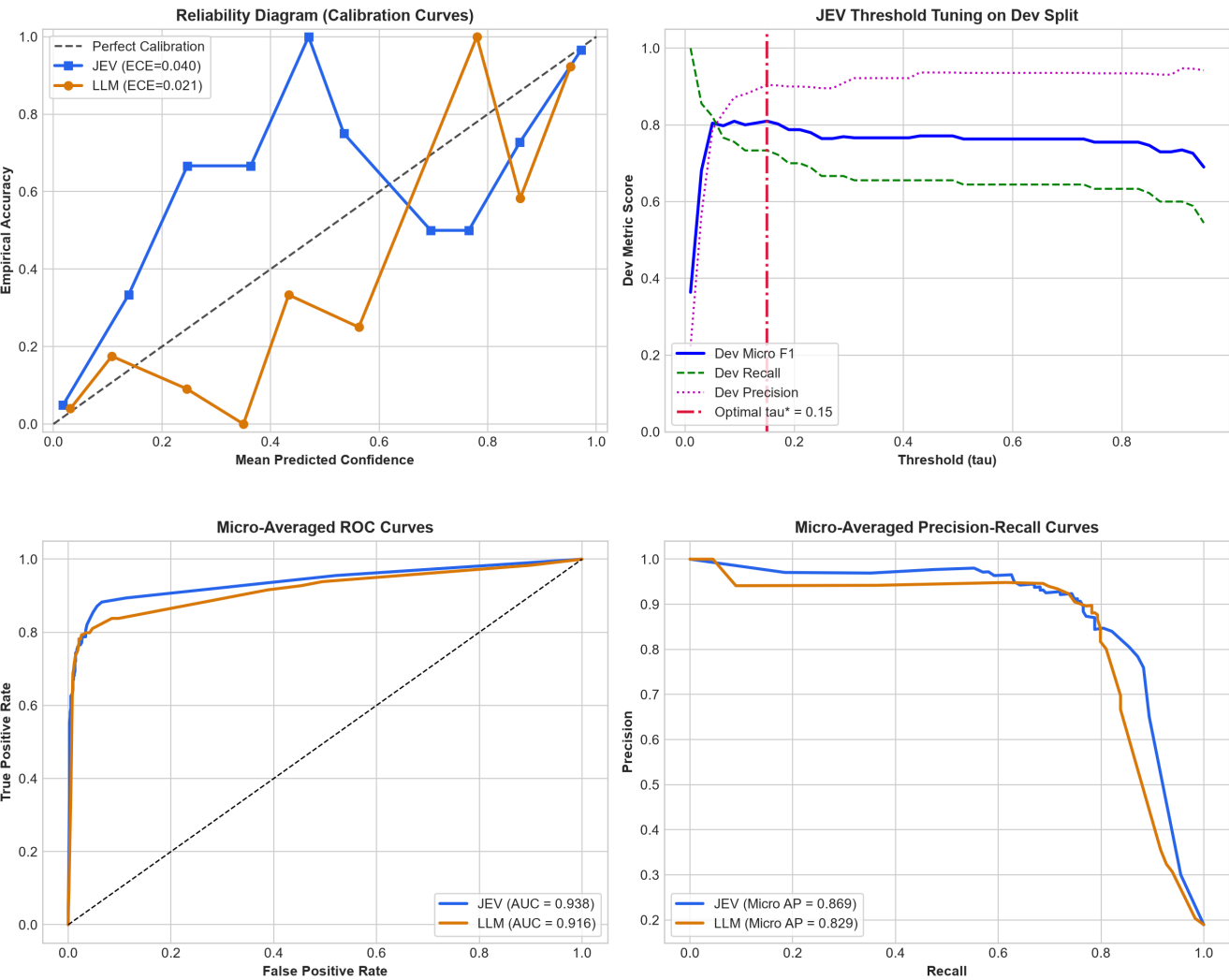
Metric	TypeSafe JEV	Groq LLM (120B)	Delta (JEV - LLM)
Overall Accuracy (All Rows)	93.7%	94.1%	-0.4%
Subset Accuracy (All 9 Match)	60.0%	59.0%	+1.0%
Micro Recall (No Misses)	0.765	0.782	-0.017
Micro Precision	0.884	0.892	-0.008
Micro F1 Score	0.820	0.833	-0.013
Macro F1 Score	0.643	0.623	+0.019
Macro Recall	0.597	0.569	+0.028
Macro Precision	0.742	0.734	+0.008
Sample Jaccard Index	0.722	0.723	-0.001
Hamming Loss (Lower=Better)	0.063	0.059	+0.004
Any-Miss Rate (Lower=Better)	34.3%	33.3%	+1.0%
False Alarm Rate (Lower=Better)	14.3%	14.3%	+0.0%



Ranking, Calibration & Reliability Diagram

Evaluation of probability calibration, Brier score loss, Expected Calibration Error (ECE), and Dev threshold tuning sweep curves.

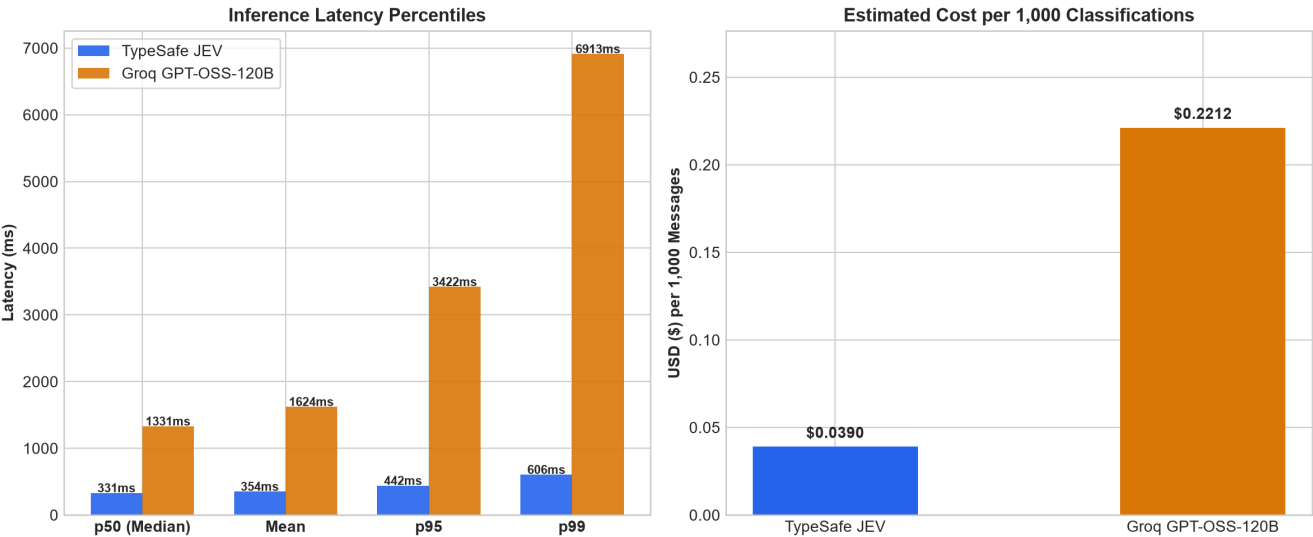
Metric	TypeSafe JEV	Groq LLM (120B)	Advantage
Micro Average Precision (mAP)	0.869	0.829	JEV
Macro Average Precision	0.683	0.636	JEV
Micro ROC-AUC	0.938	0.916	JEV
Macro ROC-AUC	0.834	0.809	JEV
Brier Score (Lower=Better)	0.0566	0.0538	LLM
Expected Calibration Error (ECE)	0.0397	0.0215	LLM



Operational Efficiency, Reliability & Latency Profiling

Runtime latency distributions, token overheads, financial cost projections per 1,000 requests, and JSON parsing reliability metrics.

Performance Dimension	TypeSafe JEV	Groq LLM (120B)	Ratio / Delta
Latency p50 (Median)	331.3 ms	1331.2 ms	4.0x Faster (JEV)
Latency Mean	353.6 ms	1624.3 ms	+1270.7 ms
Latency p95	441.6 ms	3422.0 ms	+2980.4 ms
Prompt Tokens / Request	928	499	--430 tokens
Completion Tokens / Request	157	195	-38 tokens
Cost per 1,000 Messages	\$0.0390	\$0.2212	5.7x Cheaper (JEV)
Parse Failure Rate	0.00%	0.00%	Deterministic (JEV)
Hallucinated / Extra Keys	0	0	0 Hallucinations



Statistical Significance & Genre Stratification

Paired bootstrap 95% confidence intervals, non-inferiority recall hypothesis testing, McNemar's test for exact match differences, and stratified genre performance breakdown.

Statistical Test / Metric	JEV 95% CI	LLM 95% CI	Paired Delta (95% CI)
Micro F1 Score	[0.760, 0.870]	[0.781, 0.877]	[-0.047, 0.022]
Micro Recall (Miss Penalty)	[0.696, 0.828]	[0.718, 0.842]	[-0.060, 0.030]
Micro Precision (FP Penalty)	[0.819, 0.933]	[0.838, 0.939]	[-0.057, 0.046]
Non-Inferiority Test (Recall)	Margin: -0.03	5th Pct: -0.055	Inconclusive / Inferior (Recall) (p=0.273)
Non-Inferiority Test (Precision)	Margin: -0.03	5th Pct: -0.050	Inconclusive / Inferior (Precision) (p=0.185)
McNemar Exact-Match Test	Discordant: 8 (JEV) vs 7 (LLM)	Exact Binomial Test (discordant=15) (Chi2=0.07)	p = 1.0000 (No significant difference)

